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Supporting Information for

Near-Term Suicidality Risk Encoded in the Temporal Dynamics of the Death IAT

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Software S1 to Sx

Supporting Information Text

Section 1. Assessing Uniformity of Stimulus Across Trials and Blocks In the Results section, we showed the SI status is mainly reflected in early trials right after the block switch. Specifically, this encoding is most evident in one isolated trial, e.g., trial 4 in "Death + Me" blocks, with a partial presence on encoding in adjacent trials, e.g., 3, 5, and 6. The concentration of temporal encoding in isolated trials, raises the concern on potential stimulus-specific artifacts or trigger present in the task design and in particular stimulus wording settings, pushing cognitive dynamics. To make sure this is not the case, and to exclude this possibility, we examined whether particular stimulus words were preferentially presented at these trial indexes within "Death _+ Me" blocks.

To rule out this possibility, we tested whether the composition of stimulus words differed systematically at specific trial indices. For a given trial position, we first formed the observed count vector by summing, across all participants, how many times each word was presented at that position. Let n_t denote the total number of word presentations at that trial. We then constructed a reference distribution by pooling counts of the same words across all other trial positions within the block and converting those totals into probabilities. Under the null hypothesis that stimulus allocation is homogeneous across positions, the counts at the target trial should be well described as a draw from a multinomial distribution with sample size n_t and category probabilities given by this pooled distribution.

We quantified the discrepancy between the observed counts and the expected counts (n_t multiplied by the pooled probabilities) using the Pearson chi-square goodness-of-fit statistic[1]. Because some categories have relatively small expected frequencies, we assessed significance using Monte-Carlo sampling. Specifically, we generated many synthetic trial-t count vectors by repeatedly drawing multinomial samples with the same n_t and pooled probabilities. For each synthetic sample we recomputed the chi-square statistic, thereby obtaining the distribution of mismatch values expected purely from random variation in word assignment. The Monte-Carlo p-value corresponds to the proportion of simulated statistics that were at least as large as the empirical one. Across all trial positions, including those showing the strongest temporal encoding (e.g., trial 1, 3, 4, 5, and 6) in the main analyses, we observed no significant departures from the pooled distribution.

Fig. S1A illustrates the relative frequency of each stimulus across trial index, which shows to be reasonably homogeneous allocation throughout the whole block. **Fig. S1B** shows the null distribution and the observed statistic for trial index 4. The observed value lies well within the range expected under the null hypothesis; thus, it indicates that stimulus word presentation does not account for the temporal effects observed in our analysis.

Section 2. Comparative Analysis with Sequence-based Classifier To assess whether alternative decoding or classifier models can predict SI label with the same or better performance than our proposed pipeline using RT data, we conducted a comparative analysis. For the comparison, we used a set of state-of-art classifier models ranging from simpler models including logistic regression, support vector machine (SVM)[2] and multilayer perceptron (MLP). It is worth noting that the effectiveness of these models and in particular those with a deep structure (long short-term memory (LSTM)[3], transformers[4], and large language model (LLM)–based classifier), is known to depends critically on the size and scale of the available data[5,6].

To avoid potential overfitting issues, the comparative analysis incorporated regularization, sparsity-inducing penalties where applicable, and validation-based model selection across all methods. Thus, we expect the comparison to be reasonably fair and avoid potential biases arising from limited data size or train/test settings. All baseline models were evaluated using leave-one-out cross-validation (LOOCV) at the participant level (77 folds), ensuring strict separation between training and test subjects. Performance was quantified using accuracy, balanced accuracy, F1-score, sensitivity, specificity and area under the receiver operating characteristic curve (AUC). No block-based representations, task-specific feature engineering, or label-aware preprocessing were used, contributing to our effort toward a fair comparison across methods.

Fig. S2 shows ROC curves of different classifier models, including linear classifier models operating on the average of RTs across 18 blocks. The choice of the average RT per block is consistent with block-level encoding presented in the Results section and it also helps to avoid classifiers' potential overfitting. The classifier model includes logistic regression and linear SVM, where we use L1-regularization for the model training. We additionally evaluated a multilayer perceptron (MLP) applied to the same block-averaged representation. Logistic regression performed near chance (AUC = 0.421), with similarly limited discrimination observed for the linear SVM (AUC = 0.330) and the MLP (AUC = 0.507). These results indicate that coarse block-level summaries retain insufficient information for reliable SI classification and that important discriminative structure present in the trial-level dynamics is lost relative to the bilinear representation.

We next evaluated a set of generic neural network–based classifiers trained directly on trial-level RT sequences. These models include LSTM networks, transformer encoders, and one-dimensional convolutional neural networks (1D-CNNs)[7] and a LLM embeddings with a logistic regression classifier, adapted to operate on discretized RT tokens. Overall, these sequence-aware architectures achieved moderate discrimination relative to the bilinear logistic regression model. The strongest performance among them was obtained with the LSTM, which achieved a balanced accuracy of 0.682 and an AUC of 0.712. Although this indicates that recurrent architectures can recover meaningful temporal information from the sequences, performance remained below that of the proposed approach (balanced accuracy = 0.770; AUC = 0.799) suggesting the value of incorporating task-informed structure when extracting clinically relevant temporal patterns. Complete performance metrics for all models are summarized in **Table S1**.

Section 3. Diagnostic Specificity of Temporal Features The primary analyses in the main text focused on predicting near-term SI derived from EMA. Because SI is correlated with psychiatric diagnosis, an important question is whether the temporal features identified by our framework primarily reflect general case–control differences rather than SI-specific dynamics.

To address this possibility, we repeated the full analytical pipeline using diagnostic labels, major depressive disorder (MDD) versus control (CTR) in place of SI state, active vs without active. All preprocessing, feature construction, model structure, and cross-validation procedures were kept identical to those used in the main analyses. Because the labels were the only element changed, any differences in results can be attributed to the outcome definition rather than the analytic pipeline.

At the descriptive level, the early-trial divergence observed in **Fig. 3A** of the main text was not present when participants were grouped by diagnosis as shown in **Fig. S3A**. Group-averaged trajectories showed substantial overlap, and no consistent separation emerged at the trial positions previously associated with SI-related modulation.

At the predictive level, performance was reduced when diagnostic labels were substituted for SI status. The ROC curve obtained under MDD versus control classification in **Fig. S3B** shows weaker discriminative power than that achieved for SI in **Fig. 4A** of the main text. This reduction in performance indicates that the temporal structure inferred and leveraged by the predication model to group active vs without active SI is not present to discriminate diagnostic grouping.

Together, these results suggest that the discriminative signal identified in the main analyses is more tightly linked to short-term fluctuations in SI than to broad diagnostic grouping. The findings further support our finding that the model captures a state-sensitive cognitive dynamics, rather than stable trait differences associated with psychiatric diagnosis.

Section 4. Trial-wise Group Differences in Principal Components Uncorrected p-values and FDR-adjusted q-values are shown in **Fig. S4** to illustrate the temporal distribution and relative strength of effects across trials. This figure shows trial-wise two-sample t-tests comparing PC1 and PC2 elements differences for without active and active-SI groups, derived from "Death + Me" blocks

RTs. PC1 exhibits a clear early-trial difference between groups and in particular with largest evidence at trial 4, whereas PC2 does not show a consistent group difference across trials after Bonferroni adjustment. Trial-wise effects observed for later trials toward the end of the blocks is present in both PC1 and PC2 but do not reach statistical significance after Bonferroni correction.

Figures

Fig. S1.

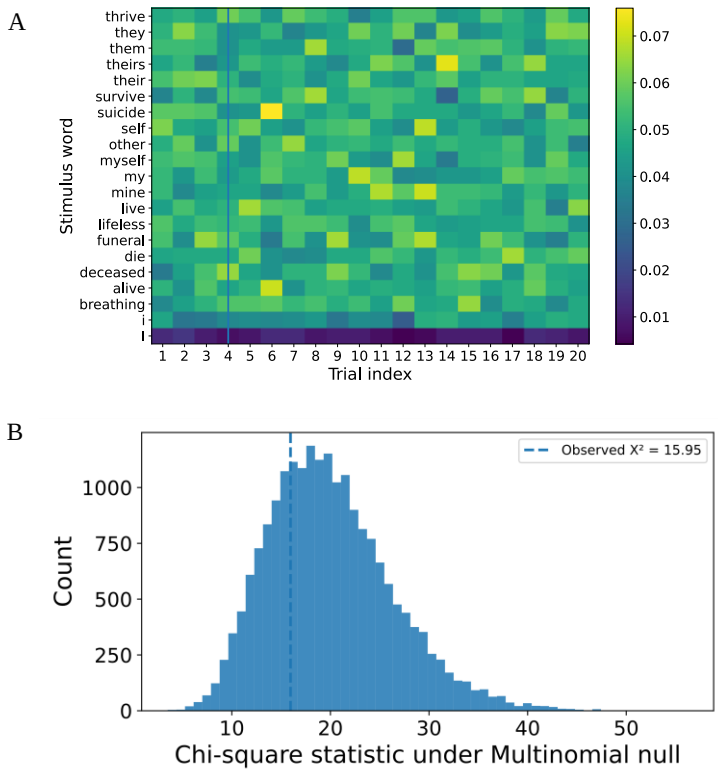

Figure S1. Control analysis for stimulus-specific effects at early trial indices within "Death + Me" blocks. (A) Heatmap showing the probability of each stimulus word as a function of trial index, computed across all participants. Colors indicate probability density. The distribution of stimulus words is reasonably uniform across trial indices, with no clear preference or lack of specific words at early trials (vertical line denotes trial 4). **(B)** Monte-Carlo evaluation of the Pearson chi-square goodness-of-fit statistic tests whether the word distribution at trial index 4 differs from the distribution pooled across all other trial indices. The histogram shows the null distribution of chi-square statistics obtained by repeatedly generating synthetic trial-4 count vectors (summed across participants) from a multinomial distribution defined by the pooled probabilities of the remaining trials and matched in total sample size. The dashed line marks the observed statistic computed from the actual trial-4 data. The observed value falls well within the null distribution, indicating that the stimulus composition at trial 4 is statistically indistinguishable from that of the remaining trials.

Fig. S2.

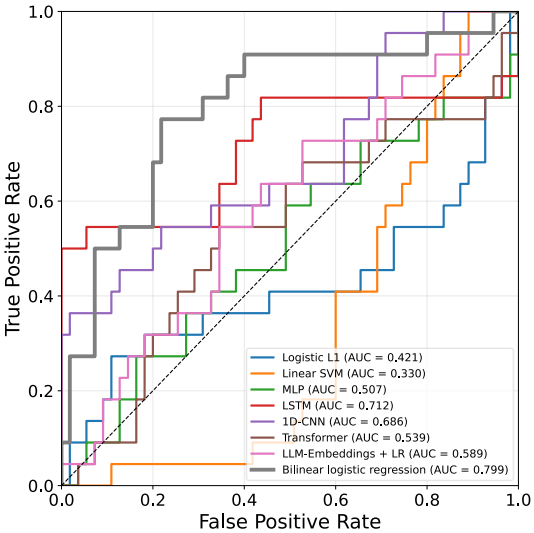

Figure S2. Performance of alternative classifiers for RT sequence decoding. Comparison of linear and neural models evaluated using participant-level leave-one-out cross-validation (77 folds). Linear classifiers applied to block-averaged representations performed near chance, indicating limited discriminative value of coarse summaries. Sequence-aware neural architectures showed improved performance, with the strongest result obtained from the LSTM (balanced accuracy = 0.682; AUC = 0.712), yet still below that achieved by the bilinear logistic regression model (balanced accuracy = 0.770; AUC = 0.799). Complete metrics for all methods are reported in Table S1.

Fig. S3.

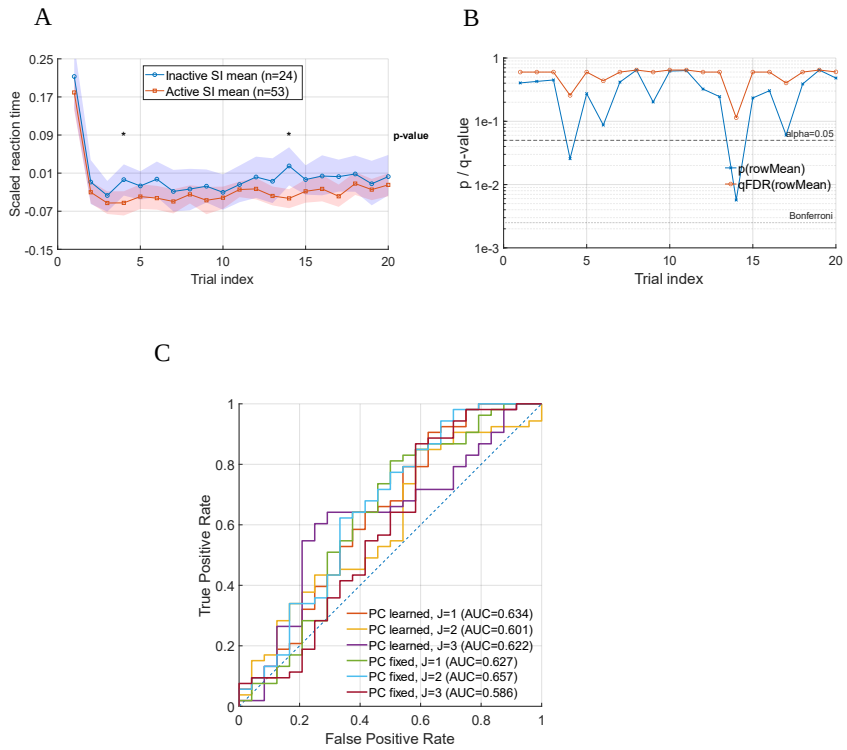

Figure S3. Evaluation of temporal features under diagnostic grouping. (A) Mean RT-scale trajectories within "Death + Me" blocks when participants are grouped by diagnostic status (major depressive disorder [MDD] versus control [CTR]). In contrast to the SI-based grouping shown in Fig. 3A of the main text, early-trial divergence is not apparent and group averages show substantial overlap across trial positions. **(B)** Mean ScaleRT is plotted as a function of trial index (1–20) for individuals with major depressive disorder (MDD) and healthy controls (CTR). For each trial position, values were first averaged across the nine repeated block instances within participant and then aggregated across subjects. Shaded regions indicate 95% bootstrap confidence intervals. The divergence between curves at specific portions of the sequence indicates that group differences are temporally localized rather than uniform across the block. **(C)** Receiver operating characteristic (ROC) curve obtained when the bilinear classification model is trained to predict diagnostic status rather than SI. Discrimination is reduced relative to SI decoding (Fig. 4A), indicating that the temporal features captured by the model are not primarily driven by case–control differences.

Fig. S4.

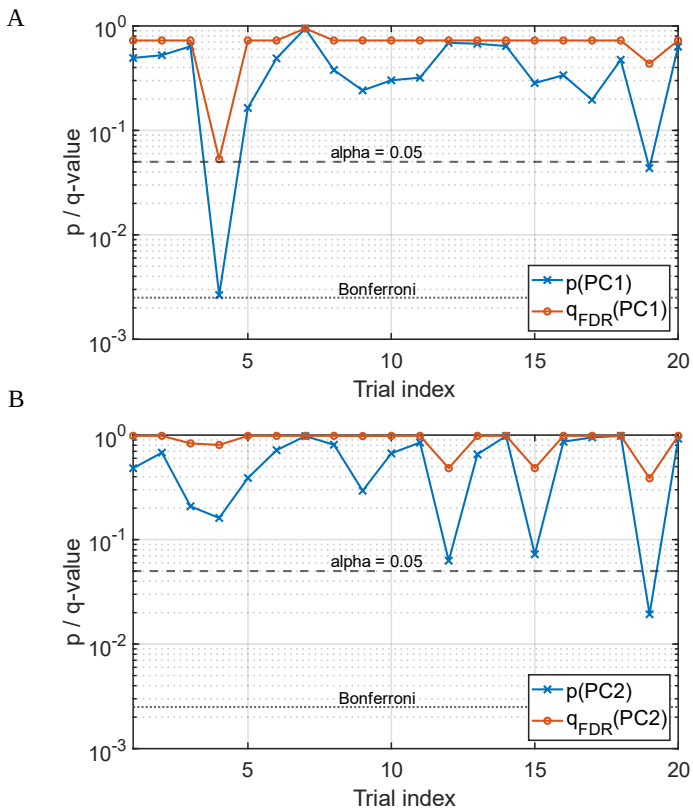

Figure S4. Trial-by-trial group comparisons of PC1 and PC2 loadings. **(A)** shows uncorrected p-value and corrected q-value for PC1 loadings derived from "Death + me" with strong encoding evidence in trial 4 where p- and q-value show statistical significance of 0.0025 and 0.05 respectively. Toward the end of block, there is evidence of encoding but it does not reach meaningful statistical significance after correction **(B)** Uncorrected p-values and FDR-adjusted q-values for PC2 loadings derived from "Death + Me" blocks. In contrast to PC1, no trial exhibits statistically significant group differences after multiple-comparison correction. In contrast to PC1, no trial shows statistically significant group differences after multiple-comparison correction. Although some trials display uncorrected effects, these do not remain statistically significant following FDR adjustment, indicating the absence of consistent encoding evidence in PC2 across the block. Blue curves show uncorrected p-values and red curves show FDR-adjusted q-values. Dashed horizontal lines indicate the nominal significance threshold (α = 0.05), and dotted lines indicate the Bonferroni-corrected threshold.

Fig. S5.

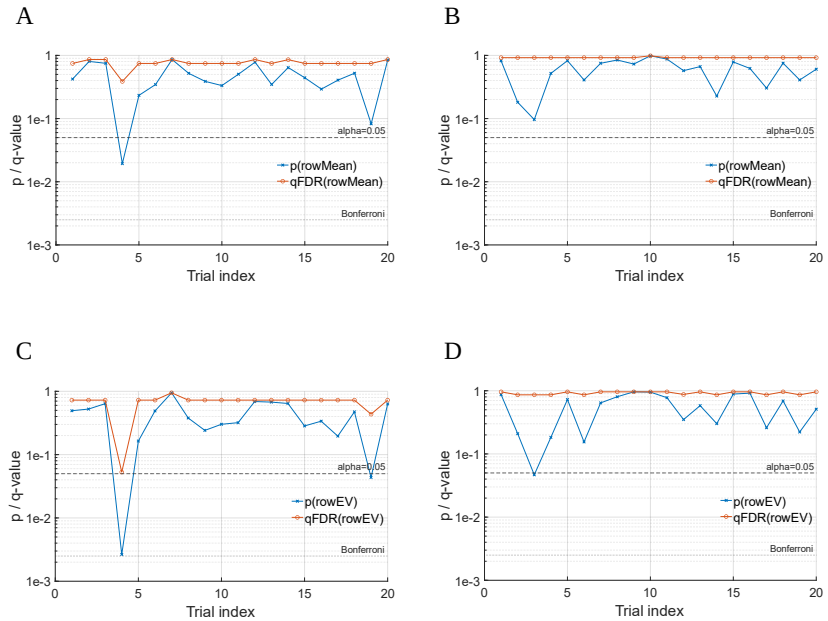

Figure S5. Trial-wise statistical comparison between active and without active SI groups. At each trial position (1–20), we performed two-sample tests with unequal variance and report both the raw p-values (×) and Benjamini–Hochberg false discovery rate–adjusted q-values. Panels **(A)** and **(B)** correspond to comparisons of the mean RT-scale in the "Death + Me" and "Life + Me" conditions, respectively. Panels **(C)** and **(D)** show the same analysis applied to the trial-wise entries of the first principal component (PC1) derived from the nonlinear transformation and uncentered PCA of the block trajectories. Horizontal dashed and dotted lines indicate the nominal significance level (α = 0.05) and the Bonferroni-corrected threshold across trial positions.

Tables

Table S1. Performance comparison between baseline classifiers and the proposed bilinear model. Classification metrics obtained under leave-one-out cross-validation (LOOCV) at the participant level. All models were trained on identical preprocessed RT sequences without access to block structure or task-informed feature engineering. Metrics reported are balanced accuracy, sensitivity, specificity, and area under the receiver operating characteristic curve (AUC). Among the baselines, the one-dimensional convolutional neural network (1D-CNN) achieved the strongest discrimination, but remained below the performance of the bilinear logistic regression model.

Model	Balanced Accuracy	F1-score	Sensitivity	Specificity	AUC
Logistic L1	0.455	0.291	0.364	0.545	0.421
Linear SVM	0.436	0.186	0.182	0.691	0.330
MLP	0.500	0.279	0.273	0.727	0.507
LSTM	0.682	0.545	0.545	0.818	0.712
1D-CNN	0.509	0.4	0.636	0.382	0.686
Transformer	0.527	0.393	0.545	0.509	0.539
LLM Embeddings with Logistic Regression Classifier	0.5455	0.355	0.364	0.727	0.589
Bilinear logistic regression	0.770	0.667	0.773	0.782	0.799

SI References

Sample References:

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